CHAPTER IV

4.2(B) How should AI systems be embedded in our social relations?

An embedded device is traditionally defined as one that incorporates a computer system to perform specialised functions and deliver real-time computing capabilities. In comparison to a general-purpose computer, it is, on one hand, "more compact". It is not general and addresses just particular needs, such as controlling specific devices or performing dedicated tasks, which limits its versatility compared to general-purpose computers. Consequently, it employs a processor with inferior performance, such as a microcontroller, and the memory is constrained by the device's dimensions. Consequently, the software system operating on the device is lightweight, typically utilising a compact embedded operating system, with the application software designed to perform specific, limited functions and reduced in scale.

With the progression of computer software and hardware technologies, embedded devices have gotten increasingly sophisticated, and their "intelligence" continues to elevate, as shown by smartphones.

The capacity for autonomous analysis and decision-making executed on embedded devices is termed 'embedded artificial intelligence'. In a specific context, embedded artificial intelligence refers to artificial intelligence used on embedded devices that is comparable to artificial intelligence utilised on general-purpose computers.

Embodied intelligence is intricately connected to embedded artificial intelligence. It may be considered a particular application of embedded artificial intelligence. It possesses clearer and more explicit specifications for embedded devices, specifically those with a physical casing, which allows for more effective integration of artificial intelligence in practical applications. In this context, embodied intelligence constitutes a subset of embedded intelligence.

Let us examine the interplay between AI and embedded systems, as well as their evolution.

Image Identification

Image Recognition. Advancements in embedded technology will facilitate the creation of imaging devices with superior processing capabilities and diminished size. Concurrently, AI will provide the necessary algorithms for real-time image and video recognition, enabling faster and more accurate identification of potential threats in various environments. The implementation of this advanced imaging technology for public safety will be beneficial, as it will detect possibly dangerous activity.

Autonomous vehicles depend on embedded systems for navigation, airbag deployment, anti-lock braking, and various other functions. Prominent technology companies like Google, Tesla,

and Uber are investing billions in research and development to create a driverless future. We will not abandon driving in the foreseeable future; it may require 10 to 15 years before autonomous vehicles become widespread on the highways.

In recent years, embedded systems have achieved a level of "intelligence". A growing quantity of smart devices is being introduced daily and weekly. Artificial intelligence in embedded systems has progressed from simplistic forms to intricate and polymorphic structures, reflecting the dynamics of biological evolution. For example, intelligent lighting systems in homes automatically activate and deactivate based on the presence of humans in the area. This system seems trivial at first glance. Upon more reflection, one acknowledges that this system independently makes decisions. The System-on-a-Chip (SoC) assesses the sensor's input to decide whether to engage the light. Isn't this a fundamental expression of AI in embedded systems?

The integration of Artificial Intelligence (AI) into human society is now a developing reality rather than a potential future. The primary challenge resides not in the incorporation of AI into our social relationships, but in its assimilation to enhance human welfare, equity, and ethical coexistence. AI systems should be included in our social interactions in a way that preserves human values, strengthens social institutions, and promotes human dignity, rather than replacing or diminishing human decision-making. This requires a human-centred, ethical, and socially responsible approach. Artificial intelligence increasingly enhances communication, decision-making, and social interaction. Therefore, AI must be included in social interactions in a human-centric and ethical manner.

AI systems should be included in our social interactions within a human-centric, ethical, and rights-based framework that prioritises human well-being, agency, and social justice. This requires a comprehensive plan that integrates technical design, policy development, and public education to ensure that AI systems are effectively aligned with ethical standards and promote equitable access for all members of society.

Integrating AI systems into social interactions requires the avoidance of emotional manipulation and coercive nudging, as these methods undermine human autonomy, dignity, and informed consent. Artificial intelligence systems can evaluate emotions, preferences, and behavioural patterns that influence human decision-making. This power, when used manipulatively, can lead individuals to make decisions they would not voluntarily choose, converting persuasion into coercion, particularly in contexts such as advertising, social media, and political campaigns where emotional manipulation is prevalent.

Human influencing AI embodied system

Virtual assistants

Virtual assistants are AI-driven systems, including Siri, Alexa, and Google Assistant, intended to engage with consumers via voice or text communication. They assist in executing duties such as responding to enquiries, establishing reminders, transmitting messages, and organising schedules, which facilitate routine human operations.

Through conversational interaction, virtual assistants serve as social intermediaries between individuals and digital services. They affect interpersonal communication, decision-making processes, and the structuring of daily activities by influencing how individuals interact with each other and with technology, potentially leading to changes in social norms and behaviours. As time progresses, users may cultivate trust or emotional dependence on these systems, prompting ethical considerations of autonomy, privacy, and diminished human interaction.

Navigation applications

Navigation applications like Google Maps and Waze employ artificial intelligence algorithms to deliver real-time directions, traffic information, and route recommendations. They facilitate users' arrivals at locations effectively by evaluating location data, traffic trends, and user behaviour.

These applications function as social intermediaries by shaping individuals' navigation through communal public areas. By guiding substantial numbers of users along specific pathways, they can alter traffic patterns, neighbourhood congestion, and even social dynamics in urban environments, potentially leading to increased reliance on these applications for navigation rather than personal exploration. Reliance on navigation applications may diminish personal spatial awareness and decision-making independence.

Unsolicited email filtering

Spam filters are artificial intelligence algorithms employed in email and messaging networks to autonomously identify and obstruct undesirable or malicious communications. They examine patterns, phrases, and sender behaviours to categorise messages as spam or legitimate. Spam filters serve as social mediators, controlling which messages users receive and which remain hidden. This influences communication by diminishing exposure to scams and harassment; yet, it may inadvertently obstruct significant communications, such as important messages from friends or family, which can lead to misunderstandings and a sense of isolation. Consequently, AI impacts trust, accessibility, and the dynamics of social interaction.

Healthcare

In healthcare, artificial intelligence systems are employed for functions including disease diagnosis, medical image analysis, patient risk prediction, and hospital resource management. These systems assist physicians and healthcare practitioners in decision-making and patient management.

Artificial intelligence in healthcare serves as a social intermediary between patients and medical practitioners. It affects diagnoses, treatment alternatives, and access to care, consequently influencing the trust dynamics between physicians and patients. Although AI enhances accuracy and efficiency, it simultaneously engenders ethical dilemmas of responsibility, bias, and the diminishment of human judgement in critical medical choices.

Social networks and digital media

Artificial intelligence is essential in influencing contemporary social networks and digital media. AI systems organise news feeds, suggest material, tailor advertisements, and oversee online interactions, significantly impacting communication, perspectives, and social connections. Social networks and digital media progressively influence communication, perspectives, and social connections. When driven by AI, these platforms shape individuals' perceptions, shared behaviours, and beliefs, rendering their ethical design imperative. Algorithms powered by artificial intelligence curate materials, suggest connections, and customise feeds. This fosters interaction but may also lead to filter bubbles, misinformation, emotional manipulation, and polarisation if it is not properly managed. Although AI enhances user experience and connectedness, it simultaneously generates ethical dilemmas, including misinformation, filter bubbles, emotional manipulation, privacy erosion, and social polarisation. Algorithmic prioritisation of participation might enhance sensational or controversial content, impacting democratic discourse and mental health by leading to the spread of misinformation and creating echo chambers that reinforce existing biases.

Social mediation denotes the role of AI as a middleman among individuals, influencing their interactions. Rather than direct human interaction, AI mediates, filters, or enhances communication. Social media algorithms, chatbots, and recommendation systems exemplify the function of AI in this capacity.

Recommendation algorithms are artificial intelligence systems engineered to suggest content, individuals, or communities based on your historical behaviour. Via social media:

Who you see: The algorithm determines which posts, videos, or stories from friends are displayed in your feed. This influences your perspective of social reality; if you are not exposed to diverse viewpoints, your social "bubble" constricts. Whom you associate with: AI selects

suggested friends, followers, or professional contacts based on shared connections, hobbies, or behavioural patterns. This influences the development and variety of your social network. What perspectives do you encounter? Content recommendations emphasise posts akin to your previous engagements. This may strengthen preexisting ideas (confirmation bias) and gently affect social attitudes or discourse.

An embodied AI system can result in Reduced social skills

Reduced social skills denote a deterioration in face-to-face communication capabilities and interpersonal interactions resulting from an augmented dependence on AI and digital technology for communication and support. Frequent involvement with AI systems—such as chatbots, virtual assistants, or social media algorithms—may result in diminished direct human interaction. This may diminish abilities such as empathy, active listening, and dispute resolution. Over time, AI-mediated communication may supplant human relationships, resulting in social isolation and shallow exchanges.

Deeper isolation

Deeper isolation denotes the heightened feeling of loneliness and social detachment that may occur when individuals depend predominantly on AI-mediated connections rather than substantive human relationships. Loneliness is most succinctly characterised as the disparity between wanted and actual social relationships. Although loneliness is a universal experience, it has become the subject of considerable speculation and commentary. As AI technologies such as chatbots, virtual assistants, and tailored social media feeds supplant or diminish human-to-human interaction, individuals may experience social withdrawal. Algorithmic personalisation may engender echo chambers, restricting exposure to a variety of social interactions. This may exacerbate feelings of isolation despite continuous digital interaction, undermining community ties and social cohesion.

Ethical embedding is the cornerstone of a healthy atmosphere.

The ethical integration of AI in social networks necessitates human oversight, openness, and accountability. Users must be educated on the influence of algorithms on their feeds and provided with substantial control over personalisation. Platforms must eschew deceptive design practices, provide informed consent, and safeguard susceptible users. Consequently, artificial intelligence in social media must preserve individual autonomy and dignity. Ethical embedding necessitates openness, enabling users to understand content selection, together with informed consent for behavioural manipulation. Platforms must eschew dark patterns and manipulative urging while guaranteeing human monitoring and accountability.

Safeguarding autonomy: Safeguarding autonomy is a fundamental ethical obligation when integrating AI systems into social interactions. Autonomy refers to the capacity of individuals to make independent, informed, and voluntary decisions on their own life. AI systems can affect decisions via recommendations, nudges, or personalisation. When appropriately developed, they facilitate decision-making; when misapplied, they undermine or distort human choice. Safeguarding autonomy guarantees that AI aids rather than governs human conduct. Individuals ought to make choices based on free will, rather than algorithmic coercion.

Upholds dignity: Human dignity entails acknowledging individuals as entities of intrinsic value, rather than just data points or objectives for enhancement. Treats individuals as moral actors rather than instruments to be optimised. AI systems that influence behaviour, exploit weaknesses, or render obscure decisions might dehumanise persons, turning them into mere variables within an algorithm. Upholding dignity guarantees that AI acknowledges individuals as moral actors endowed with reason, choice, and autonomy.

Preserving trust is a vital rationale for integrating AI technologies within the frameworks of human oversight, transparency, and ethical governance. Trust influences the willingness of individuals and society to adopt and depend on AI technologies. AI systems that function transparently, honour autonomy, and stay under human oversight are more likely to be regarded as equitable, dependable, and responsible. Awareness of human oversight in AI decision-making and the ability to rectify errors enhances confidence in the system. In contrast, opaque, manipulative, or entirely autonomous AI systems undermine trust by instilling fear, uncertainty, and perceptions of inequity. Ethical precautions, including explainability, informed consent, and accountability, are essential for maintaining public trust.

- Human agency and overarching supervision are crucial for the ethical integration of AI systems inside society. Human agency guarantees that individuals maintain control, choice, and decision-making authority, whereas general supervision maintains ongoing human monitoring of AI operations. AI systems ought to serve as decision-support instruments rather than autonomous decision-makers, particularly in critical fields such as healthcare, law, and governance. Humans must possess the capability to monitor, intervene, overrule, or deactivate AI systems when required. Preserving human agency guarantees moral and legal accountability, as only individuals can be held responsible for decisions and their repercussions. It safeguards human dignity and autonomy, averting uncritical dependence on algorithmic results. In high-stakes contexts, such as healthcare and criminal justice, artificial intelligence systems should serve as instruments that enhance human decision-making rather than supplant it. The ultimate responsibility and accountability must always reside with a

human being. Maintaining human accountability for crucial decisions is a fundamental principle for the ethical integration of AI into society. It guarantees that ethical, legal, and critical decisions are maintained within human discretion, rather than being wholly assigned to algorithms. Artificial intelligence can analyse extensive datasets and recommend ideal behaviours; nevertheless, it is devoid of moral agency, empathy, and accountability. Delegating crucial choices exclusively to AI may result in errors, injustice, or injury. Humans must maintain supreme authority, particularly in matters that directly affect life, liberty, and welfare.

- The principles of non-discrimination and equity need the development and training of systems to mitigate the effects of algorithmic bias. This will guarantee that the systems do not perpetuate or exacerbate existing societal biases based on factors such as age, gender, ethnicity, or other characteristics. The outcomes must be equitable for all various groups.
- Transparency and Comprehensibility: The decision-making processes of artificial intelligence, especially those affecting individuals' lives, must be understandable and interpretable by people. Trust is established as a consequence, allowing individuals to interrogate decisions.
- Confidentiality and Data Security: Confidentiality and data security pertain to the safeguarding of personal, sensitive, and private information gathered and processed by AI systems. In sectors such as healthcare, social media, and virtual assistants, AI depends on enormous quantities of user data to operate well. When AI systems are integrated into social relationships, they frequently obtain personal information regarding individuals' lives, activities, and behaviours. Inadequate security protocols or data mismanagement can result in privacy violations, surveillance, or unauthorised dissemination of information. This erodes confidence between individuals and institutions and poses ethical issues regarding consent, control, and accountability.

Legislation, including the General Data Protection Regulation (GDPR), mandates the implementation of stringent processes throughout the lifecycle of artificial intelligence to safeguard personal data and uphold individuals' right to privacy.

- Accountability: When an artificial intelligence system inflicts harm or commits an error, it is essential to delineate explicit lines of responsibility to ascertain who is liable for the incident (such as the developer, the deployer, or the user organisation).
- Security and Resilience: Artificial intelligence systems must be technically robust, reliable, and resilient against malicious attacks. This will guarantee that they do not inflict any unforeseen damage or disruption to the system. Security and resilience in AI pertain to safeguarding AI systems from cyberattacks, misuse, and operational failures, along with their

capacity to recuperate and maintain functionality in bad circumstances. The integration of AI systems into essential social infrastructures, including healthcare, transportation, communication, and government, renders them susceptible to security breaches or system failures that may result in extensive societal damage. Inadequate resilience may result in service interruptions, decision manipulation, and erosion of public trust. Consequently, guaranteeing robust and resilient AI systems is vital for social stability and safety.

Design and Development Guided by Ethical Principles

• Ethics-by-Design denotes the methodology of embedding ethical principles—such as justice, transparency, accountability, privacy, and safety—into the design and development of AI systems, rather than confronting ethical concerns post-deployment. Integrating ethics into algorithms, data selection, and system architecture enables developers to avert issues such as bias, discrimination, privacy infringements, and misuse. Ethics-by-Design guarantees that AI systems integrated into social interactions conform to human values and societal standards. This proactive strategy is essential, as rectifying ethical breaches becomes challenging and expensive as AI systems influence social interactions. Conversely, it is imperative that ethical principles be included in the artificial intelligence development process from the outset, rather than as an afterthought.

•Diverse Teams: Diverse teams denote AI development groups comprised of persons from various backgrounds, encompassing multiple genders, cultures, disciplines, and experiential perspectives. This diversity facilitates the incorporation of many perspectives in the design and implementation of AI systems.

Maintaining human accountability for pivotal decisions is a fundamental principle for the ethical integration of AI into society. It guarantees that ethical, legal, and critical decisions are maintained within human discretion, rather than being wholly assigned to algorithms. AI can analyse extensive datasets and recommend the best behaviours; nevertheless, it is devoid of moral agency, empathy, and accountability. Delegating crucial choices exclusively to AI may result in errors, injustice, or injury. Humans must maintain supreme authority, particularly in matters that directly affect life, liberty, and welfare. Development teams must be diverse to identify and rectify potential biases present in algorithms and training data. This will facilitate the identification and rectification of any existing biases.

The integration of AI into social connections, coupled with a deficiency of diversity among designers, can lead to biased systems that neglect or penalise specific groups. Diverse teams are more proficient at recognising ethical hazards, cultural sensitivities, and potential social

repercussions early in the design process. This results in more inclusive, equitable, and socially accountable AI systems.

Ethicists and social scientists are specialists who examine moral ideals, social behaviour, power dynamics, and cultural environments. In AI development, they assist in analysing the impact of AI systems on individuals, communities, and society as a whole. When AI is integrated into social interactions, mere technological efficiency is inadequate. Ethicists and social scientists recognise ethical dangers, including bias, inequity, loss of autonomy, and social marginalisation. Their participation guarantees that AI systems adhere to human values, social conventions, and democratic ideals, augmenting the efforts of engineers and data scientists.

Incorporating individuals of all genders, cultures, and age demographics in AI development teams ensures the representation of a wide array of life experiences, social contexts, and viewpoints in the design and implementation of AI systems. AI systems integrated into social relationships impact various people in distinct ways. In the absence of diversity within development teams, AI may embody limited perspectives and perpetuate stereotypes or exclusions. Diverse representation facilitates the recognition of cultural biases, gender-specific effects, and generational disparities in technology utilisation, resulting in more equitable and inclusive AI systems.

• Impact Assessments: Impact assessments are methodical studies conducted prior to and throughout the implementation of AI systems to ascertain their potential social, ethical, legal, and economic ramifications. They seek to foresee the risks and detriments that AI may inflict on individuals or society. When AI systems are integrated into social relations, their decisions can influence rights, opportunities, and trust. Impact evaluations facilitate the identification of issues, including bias, discrimination, privacy infringements, and security vulnerabilities, proactively. This proactive strategy enables developers and regulators to adjust or limit AI systems prior to inflicting widespread societal damage.

To proactively identify and mitigate adverse consequences, it is essential to conduct mandatory Ethical and Societal Impact Assessments (EIA/SIA) before using artificial intelligence in socially sensitive domains. This is essential to eliminate undesirable effects.

Collaboration between humans and artificial intelligence should be promoted. Artificial intelligence should serve as an instrument that enhances human intellect rather than supplanting it. Collaboratively, humans and AI can tackle intricate global issues such as climate change, healthcare enhancement, and economic advancement.

Consequently, integrating AI into our social interactions necessitates a measured strategy that harmonises scientific advancement with ethical accountability. When informed by human values, democratic ideals, and social justice, AI can serve as a formidable catalyst for beneficial societal progress.

Knock your brain

1Embedding ethics into AI design from the start is called:

A) Impact Assessment

B) Security and Resilience

C) Ethics-by-Design

D) Social Mediation

2 Excessive reliance on AI for social interactions may lead to:

A) Increased human empathy

B) Reduced social skills and deeper isolation

C) Improved face-to-face communication

D) Greater diversity in teams

3 What does embedding AI in social relations primarily involve?

A) Improving AI processing speed

B) Limiting AI to technical tasks only

C) Reducing electricity consumption of AI systems

D) Influencing human interactions and decision-making

4 Conducting an evaluation of potential social, ethical, and legal consequences before deploying AI is known as:

A) Impact Assessment

B) Social Mediation

C) Data Mining

D) Ethics-by-Design

5 Including ethicists and social scientists in AI development helps:

A) Improve technical coding speed

B) Identify ethical risks and social consequences of AI

C) Reduce the cost of AI hardware

D) Automate human judgment entirely

Unleash your potential

1.Analyse the influence of artificial intelligence on interpersonal relationships and societal frameworks.

2.Can artificial intelligence systems supplant human decision-making in social institutions? Engage in discourse.

3.Assess the significance of ethical norms in the incorporation of AI into society.